

Sprint 5 Submission - PetAlert (Team Sunrise Inc)

GitHub Repository: <https://github.com/SunriseInc-Capstone/PetAlert>

Team Information and Introduction

Team Name

Sunrise Inc

Project Name

PetAlert

Team Members

Ojaswi Subedi, Tanvi Biswal, Kimberley Juarez, Anjali Fernando, Bibek Pandey

Overview

Problem Context

Pet owners face confusion and lost time when a pet goes missing. Information such as photos, contacts, and identification details is often scattered across different apps or papers, making responses slower and less organized.

Goal

PetAlert aims to provide a cloud-enabled mobile application that securely stores pet profiles, emergency contacts, and alert templates in one centralized location using Firebase services. The system ensures that pet information is persistent, accessible across devices, and quickly retrievable. With a single tap, users can generate and share missing pet alerts via social media, email, SMS, or printable formats, helping reunite pets and families more efficiently.

Requirements

Functional Requirements (User Stories)

- R1 Pet Profile: Create/update pet profiles with photo and details (local storage).
- R2 Emergency Contacts: Add/edit vet, shelter, police contacts with one-tap call/email/SMS.
- R3 Missing Pet Alert: Generate social media post, email/SMS draft, and phone script instantly.
- R4 Regular Update Prompts: Local notifications reminding owners to update info/photos.
- R5 Photo Gallery: Upload/view multiple pet photos locally.
- R6 Recommendations: Static safety tips and search strategies.
- R7 Toxic Foods Infographics: Quick visual guides for dangerous foods.

Non-Functional Requirements

- NF1 Security: All data stored locally; optional OS-level lock (FaceID/Passcode).
- NF2 Reliability: Core features work offline (profiles, contacts, alerts).
- NF3 Usability: ≤3 taps to reach any function; emergency-first button on home.
- NF4 Accessibility: Support large text, high contrast, screen reader labels.
- NF5 Cross-Platform: Runs consistently on iOS 14+ and Android 10+.
- NF6 Accuracy: Generated alerts match stored details exactly.

Design

The app uses a tab-based navigation system:

Home: Prominent “Missing Pet Alert” button.

Pets: Displays all pet profiles with add/edit options.

Pet Detail: Shows stored information and allows generating an alert.

Contacts: Emergency contacts with instant action buttons.

Toxic Foods: Scrollable infographic cards and quick-reference visuals.

Reflection Report

Accomplishments

Throughout Sprint 5, we added numerous key features that increased the app's productivity. We created a reminder feature that allows users to add, modify, delete, and manage reminders for their pets. People can also select specific dates and times for reminders, which are linked to the specific pet. We also implemented the Tips & Toxic Food function, which offers users with critical safety information and emergency suggestions for their pet's health. We also increased system integration by integrating the contacts module with missing alerts, allowing users to select saved contacts rather than manually inputting them. This will decrease duplicate data entry and enhance the user experience.

What Went Well

Sprint 5 went well because our team finished several key features and improved the app's overall usefulness. We were able to build the Reminders feature, finish the Tips & Toxic Foods functionality, and improve the integration of contacts and missing tasks on time. Our team communicated effectively, attended meetings on time, and completed their assigned work. The project's structure and design grew more organized, making the app easier to maintain and extend.

What Could Be Improved

While Sprint 5 was successful, there are still areas we can improve. Some features, such as the reminders system, currently use local storage and can be improved by integrating them fully with the backend. We can also strengthen security by adding proper Firestore security rules and continue refining the user interface through additional testing to improve reliability and user experience.

Plans for Sprint 6

In Sprint 6, we intend to continue enhancing the app by integrating additional changes to enhance its functionality and reliability. This includes creating backend security rules and switching the reminders functionality from local storage to the backend for improved data management. We'll also start brainstorming ideas and planning the design of our project poster.

Rubric Reflection

What went well in the presentation?

The Sprint 5 presentation went well since we clearly presented our progress, such as finishing significant features like the reminders system and toxic food safety guidance. We also demonstrated good system integration by linking contacts to missing alerts. Our team also described and demonstrated the technical implementation, backend architecture, and individual responsibilities throughout the demo, highlighting both the functionality and design of our app.

What can be improved in the presentation?

The entire presentation ran smoothly, and everyone communicated properly. Each team member clearly discussed their role in the project, allowing the professor to better grasp the features and improvements achieved in Sprint 5.

Other Notes

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